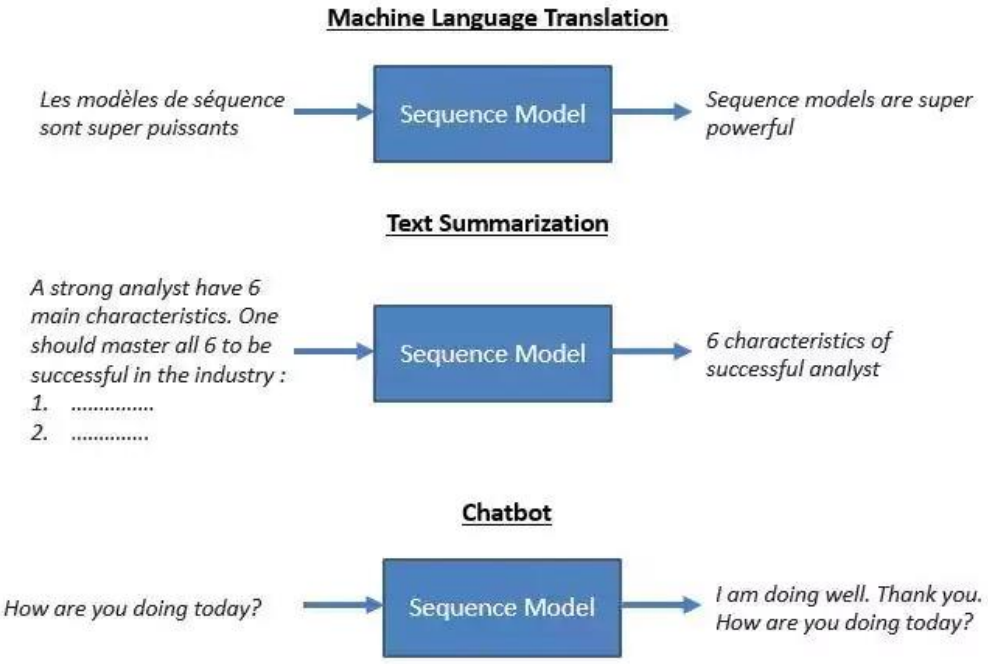
ANN and DL (CAI-361)

Objectives:

- Seq2Seq modeling
- Encoder Decoder

Seq2Seq Modeling:

Sequence-to-Sequence (Seq2Seq) problem is a special class of Sequence Modelling Problems in which both, the input and the output is a sequence. Seq2Seq models are a type of neural network, an exceptional Recurrent Neural Network architecture, designed to transform one data sequence into another. They are handy for tasks where the input and output are sequences of varying lengths, which traditional neural networks struggle to handle, such as solving complex language problems like machine translation, question answering, creating chatbot, text summarization, etc. This was the motivation behind coming up with an architecture that can solve general sequence-to-sequence problems and so **encoder-decoder** models were born.



The Encoder-Decoder architecture is relatively new and had been adopted as the core technology inside Google's translate service in late 2016. It forms the basis for advanced sequence-to-sequence models like Attention models, GPT Models, Transformers, and BERT.

Sequence Modeling Problems

Sequence Modeling problems refer to the problems where either the input and/or the output is a sequence of data (words, letters…etc.). Consider a very simple problem of predicting whether a

movie review is positive or negative. Here our input is a sequence of words and output is a single number between 0 and 1. If we used traditional DNNs, then we would typically have to encode our input text into a vector of fixed length using techniques like BOW, Word2Vec, etc. But note that here the sequence of words is not preserved and hence when we feed our input vector into the model, it has no idea about the order of words and thus it is missing a very important piece of information about the input.

Thus to solve this issue, RNNs came into the picture. In essence, for any input $X = (x_0, x_1, x_2, \dots x_t)$ with a variable number of features, at each time-step, an RNN cell takes an item/token $\mathbf{x_t}$ as input and produces an output $\mathbf{h_t}$ while passing some information onto the next time-step. These outputs can be used according to the problem at hand.

The movie review prediction problem is an example of a very basic sequence problem called many to one prediction. There are different types of sequence problems for which modified versions of this RNN architecture are used.

The Architecture of Encoder-Decoder models

This first Encoder-Decoder model was applied to English to French translation. In this model the input is a series of words, processed one after another. The output is, likewise, a series of words.

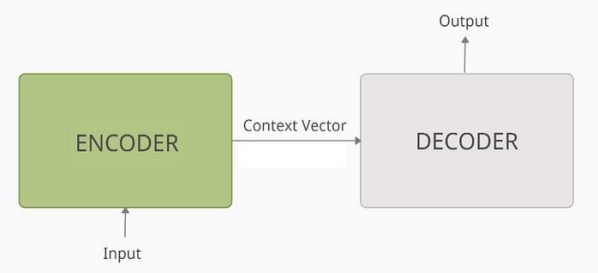
Task: Predict the French translation for every English sentence used as input.

For simplicity let's consider that there is only a single sentence in our data corpus. So let our data be:

- **Input:** English sentence: **"nice to meet you"**
- **Output:** French translation: **"ravi de vous rencontrer"**

- I will refer to the Input sentence **"nice to meet you"** as **X / input-sequence**
- I will refer to the output sentence **"ravi de vous rencontrer"** as **Y_true / target-sequence.** This is what we want our model to predict (the ground truth).
- I will refer to the predicted output sentence of the model as **Y_pred / predicted-sequence**
- The individual words of the English and French sentence are referred to as **tokens**
  Hence, given the input-sequence "nice to meet you", we want our model to predict the target-sequence/Y_true i.e "ravi de vous rencontrer"

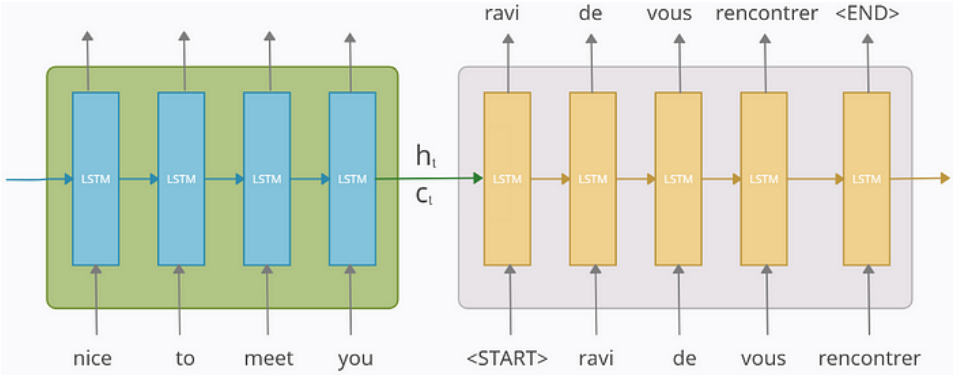
At a very high level, an encoder-decoder model can be thought of as two blocks, the **encoder** and the **decoder** connected by a vector which we will refer to as the **'context vector'.**



- **Encoder**: The encoder processes each token in the input-sequence. It tries to cram all the information about the input-sequence into a vector of fixed length i.e. the 'context vector'. After going through all the tokens, the encoder passes this vector onto the decoder.

- **Context vector**: The vector is built in such a way that it's expected to encapsulate the whole meaning of the input-sequence and help the decoder make accurate predictions. This is the final internal states of our encoder block.
- **Decoder**: The decoder reads the context vector and tries to predict the target-sequence token by token.

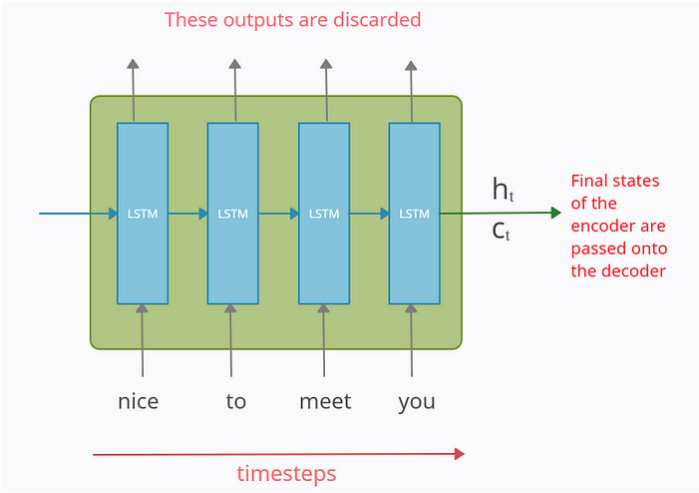
The internal structure of both the blocks would look something like this:



The model can be thought of as two LSTM cells with some connection between them. The main thing here is how we deal with the inputs and the outputs.

The Encoder Block

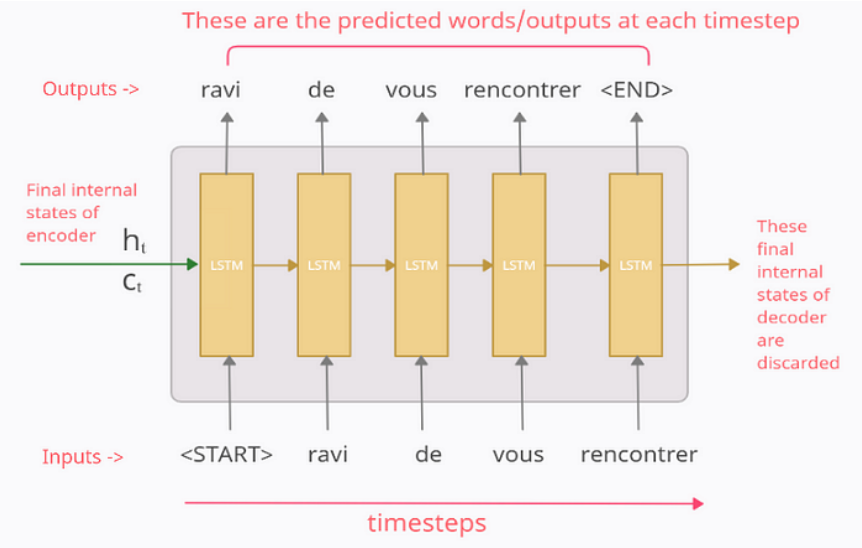
The encoder part is an LSTM cell. It is fed in the input-sequence over time and it tries to encapsulate all its information and store it in its final internal states $\mathbf{h_t}$ (hidden state) and $\mathbf{c_t}$ (cell state). The internal states are then passed onto the decoder part, which it will use to try to produce the target-sequence. This is the **'context vector'** which we were earlier referring to. The outputs at each time-step of the encoder part are all discarded



Note: Above diagram is what an LSTM/GRU/RNN cell looks like when we unfold it on the time axis. i.e. it is the single LSTM/GRU/RNN cell that takes a single word/token at each timestamp.

The Decoder Block

So after reading the whole input-sequence, the encoder passes the internal states to the decoder and this is where the prediction of output-sequence begins.



The decoder block is also an LSTM cell. The main thing to note here is that the initial states **(h_0, c_0)** of the decoder are set to the final states **(h_t, c_t)** of the encoder. These act as the 'context' vector and help the decoder produce the desired target-sequence.

Now the way decoder works, is, that its output at any time-step **t** is supposed to be the **t^{th}** word in the target-sequence/Y_true ("ravi de vous rencontrer"). To explain this, let's see what happens at each time-step.

At time-step 1

The input fed to the decoder at the first time-step is a special symbol **"<START>"**. This is used to signify the start of the output-sequence. Now the decoder uses this input and the internal states **(h_t, c_t)** to produce the output in the 1st time-step which is supposed to be the 1st word/token in the target-sequence i.e. **'ravi'**.

At time-step 2

At time-step 2, the output from the 1st time-step **"ravi"** is fed as input to the 2nd time-step. The output in the 2nd time-step is supposed to be the 2nd word in the target-sequence i.e. **'de'**

And similarly, the output at each time-step is fed as input to the next time-step. This continues till we get the "<END>" symbol which is again a special symbol used to mark the end of the output-sequence. The final internal states of the decoder are discarded.

Note that these special symbols need not necessarily be "<START>" and "<END>" only. These can be any strings given that these are not present in our data corpus so the model doesn't confuse them with any other word.

Training and Testing Phase:

Vectorizing our data

Before getting into details of it, we first need to vectorize our data.

The raw data that we have is

- **X = "nice to meet you" → Y_true = "ravi de vous rencontrer"**

Now we put the special symbols "<START>" and "<END>" at the start and the end of our target-sequence

- **X = "nice to meet you" → Y_true = "<START> ravi de vous rencontrer <END>"**

Next the input and output data is vectorized using one-hot-encoding. Let the input and output be represented as

- **X = (x1, x2, x3, x4) → Y_true = (y0_true, y1_true, y2_true, y3_true, y4_true, y5_true)**

where xi's and yi's represent the vectors for input-sequence and output-sequence respectively. They can be shown as:

For input X

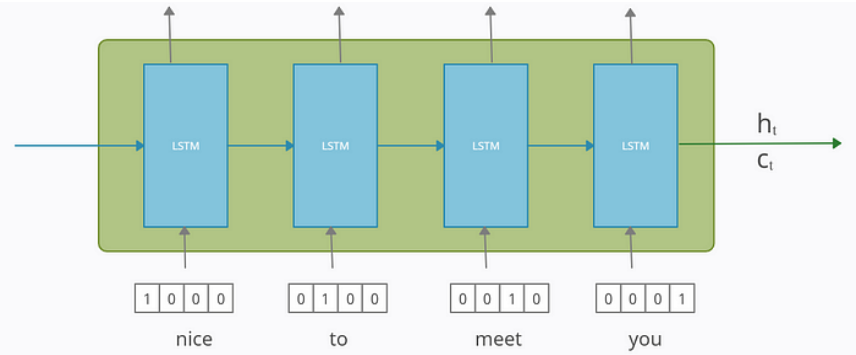
'nice' → x1 : [1 0 0 0]
'to' → x2 : [0 1 0 0]
'meet' →x3 : [0 0 1 0]
'you' → x4 : [0 0 0 1]

For Output Y_true

'<START>' → y0_true : [1 0 0 0 0 0]
'ravi' → y1_true : [0 1 0 0 0 0]
'de' → y2_true : [0 0 1 0 0 0]
'vous' → y3_true : [0 0 0 1 0 0]
'rencontrer' → y4_true : [0 0 0 0 1 0]
'<END>' → y5_true : [0 0 0 0 0 1]

Training and Testing of Encoder

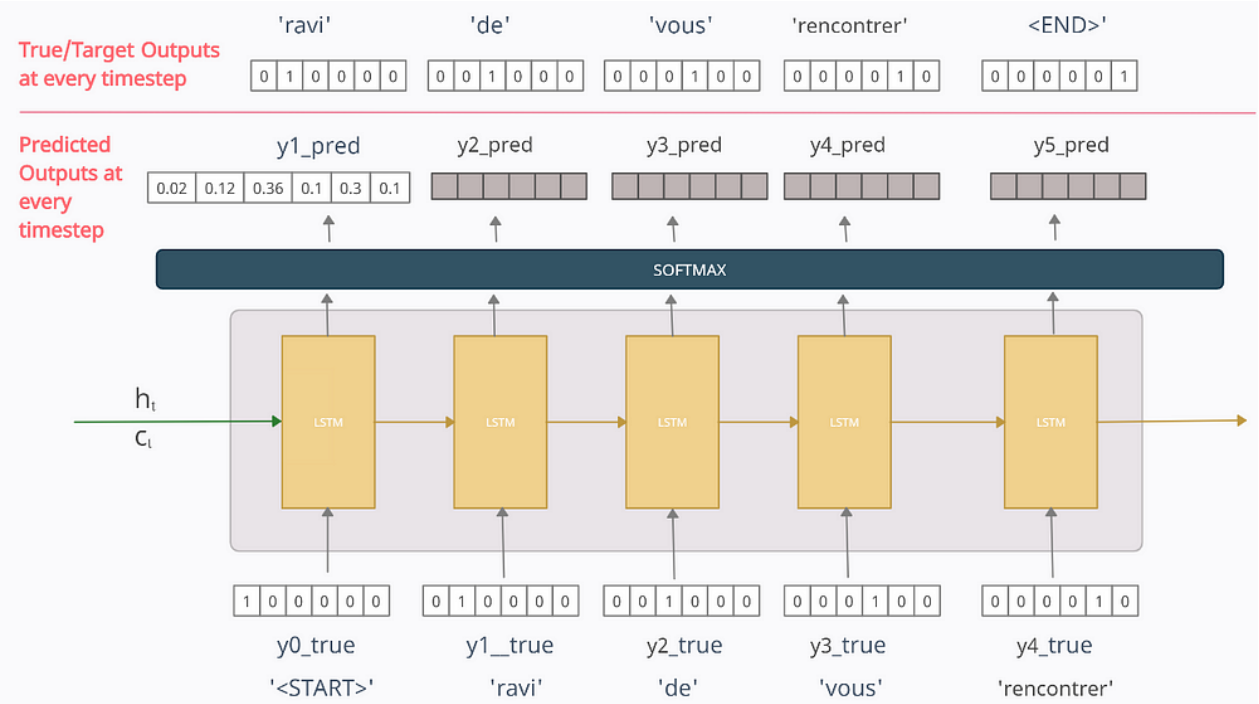
The working of the encoder is the same in both the training and testing phase. It accepts each token/word of the input-sequence one by one and sends the final states to the decoder. Its parameters are updated using backpropagation over time.



The Decoder in Training Phase:

The working of the decoder is different during the training and testing phase, unlike the encoder part. To train our decoder model, we feed the **true output / token** (and **not the predicted** output/token) from the previous time-step as input to the current time-step.

To explain, let's look at the 1st iteration of training. Here, we have fed our input-sequence to the encoder, which processes it and passes its final internal states to the decoder. Now for the decoder part, refer to the diagram below.



Before moving on, note that in the decoder, at any time-step **t**, the output **yt_pred** is the probability distribution over the entire vocabulary in the output dataset which is generated by using the **Softmax** activation function. The token with the maximum probability is chosen to be the predicted word.

For example referring to the above diagram, y1_pred = [0.02 0.12 0.36 0.1 0.3 0.1] tells us that our model thinks that the probability of 1st token in the output-sequence being '<START>' is 0.02, 'ravi' is 0.12, 'de' is 0.36 and so on. We take the predicted word to be the one with the highest probability. Hence here the predicted word/token is **'de'** with a probability of 0.36

At time-step 1

The vector [1 0 0 0 0 0] for the word '<START>' is fed as the input vector. Now here I want my model to predict the output as y1_true=[0 1 0 0 0 0] but since my model has just started training, it will output something random. Let the predicted value at time-step 1 be y1_pred=[0.02 0.12 0.36 0.1 0.3 0.1] meaning it predicts the 1st token to be **'de'**. Now, should we use this y1_pred as the input at time-step 2?. We can do that, but in practice, it was seen that this leads to problems like slow convergence, model instability, and poor skill which is quite logical if you think.

To rectify this isse we feed the true output/token (and not the predicted output) from the previous time-step as input to the current time-step. That means the input to the time-step 2 will be y1_true=[0 1 0 0 0 0] and not y1_pred.

Now the output at time-step 2 will be some random vector y2_pred. But at time-step 3 we will be using input as y2_true=[0 0 1 0 0 0] and not y2_pred. Similarly at each time-step, we will use the true output from the previous time-step.

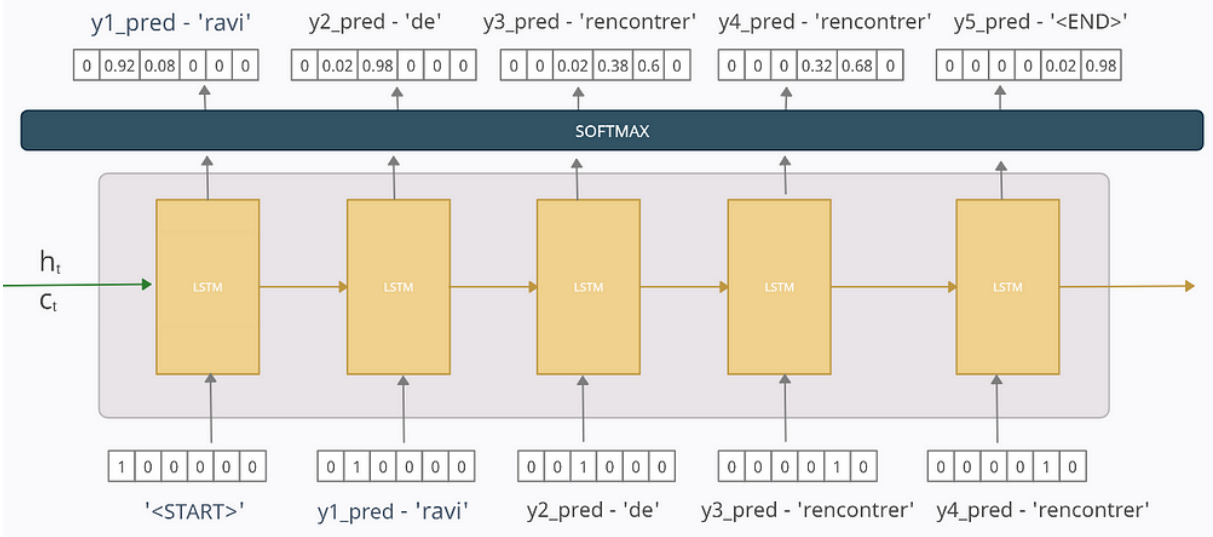
Finally, the loss is calculated on the predicted outputs from each time-step and the errors are backpropagated through time to update the parameters of the model. The loss function used is the categorical cross-entropy loss function between the target-sequence/**Y_true** and the predicted-sequence/**Y_pred** such that

- Y_true = [y0_true, y1_true, y2_true, y3_true, y4_true, y5_true]
- Y_pred = [‘<START>’, y1_pred, y2_pred, y3_pred, y4_pred, y5_pred]

The final states of the decoder are discarded

The Decoder in Test Phase

In a real-world application, we won't have Y_true but only X. Thus we can’t use what we did in the training phase as we don’t have the target-sequence/Y_true. Thus when we are testing our model, the predicted output (and not the true output unlike the training phase) from the previous time-step is fed as input to the current time-step. Rest is all same as the training phase.



At time-step 1

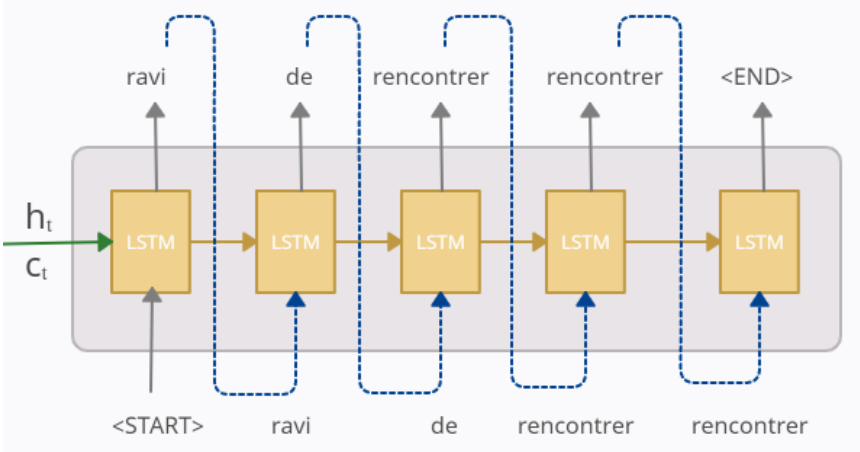
y1_pred = [0 0.92 0.08 0 0 0] tells that the model is predicting the 1st token/word in the output-sequence to be ‘ravi’ with a probability of 0.92 and so now at the next time-step this predicted word/token will only be used as input.

At time-step 2

The predicted word/token **“ravi”** from 1st time-step is used as input here. Here the model predicts the next word/token in the output-sequence to be **‘de’** with a probability of 0.98 which is then used as input at time-step 3

And the similar process is repeated at every time-step till the ‘<END>’ token is reached

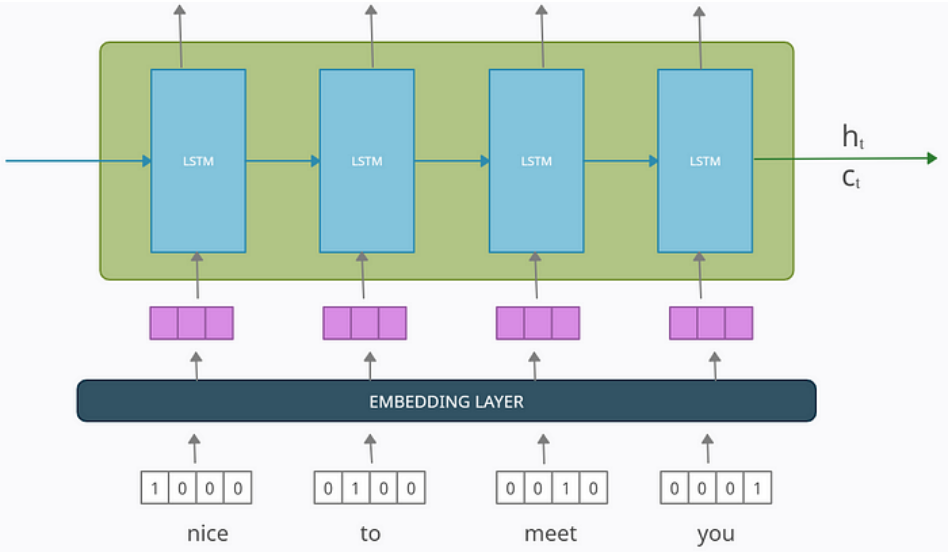
Better visualization for the same would be:



So according to our trained model, the predicted-sequence at test time is "ravi de rencontrer rencontrer". Hence though the model was incorrect on the 3rd prediction, we still fed it as input to the next time-step. The correctness of the model depends on the amount of data available and how well it has been trained. The model may predict a wrong output but nevertheless, the same output is only fed to the next time-step in the test phase.

The Embedding Layer

The input-sequence in both the decoder and the encoder is passed through an embedding layer to reduce the dimensions of the input word vectors because in practice, the one-hot-encoded vectors can be very large and the embedded vectors are a much better representation of words. For the encoder part, this can be shown below where the embedding layer reduces the dimensions of the word vectors from four to three.



This embedding layer can be pre-trained like Word2Vec embeddings or can be trained with the model itself.

The Final Visualization at test time